

VLANs, STP Configuration and Troubleshooting

VLAN issues
and
STP Configuration /
Issues

Overview of VLANs

Types of VLANs

- Default VLAN – all switch ports become part of this VLAN until switch is configured, **show vlan brief**
- Native VLAN – **used for untagged traffic (The native VLAN is the only VLAN which is not tagged in a trunk, in other words, native VLAN frames are transmitted unchanged.)**
- Management VLAN – used to access management capabilities
 - A good security practice is to separate management and user data traffic. The management VLAN, which is VLAN 1 by default, should be changed to a separate, distinct VLAN. To communicate remotely with a Cisco switch for management purposes, the switch must have an IP address configured on the management VLAN. A management VLAN is typically dedicated to managing devices.

VLAN 1

```
Switch# show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gi0/1, Gi0/2
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

- All ports assigned to VLAN 1 by default.
- Native VLAN is VLAN 1 by default.
- Management VLAN is VLAN 1 by default.

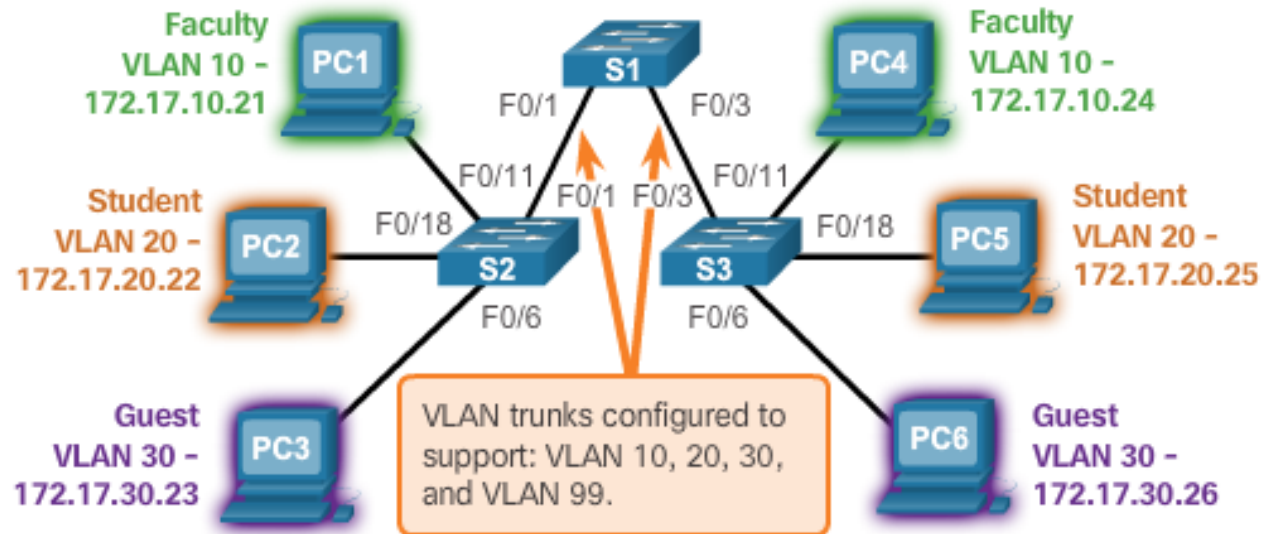
VLANs in a Multi-Switched Environment

VLAN Trunks

VLAN 10 Faculty/Staff - 172.17.10.0/24
VLAN 20 Students - 172.17.20.0/24
VLAN 30 Guest - 172.17.30.0/24
VLAN 99 Management and Native - 172.17.99.0/24

F0/1-5 are 802.1Q trunk interfaces with native VLAN 99.
F0/11-17 are in VLAN 10.
F0/18-24 are in VLAN 20.
F0/6-10 are in VLAN 30.

Example network:



The links between switches S1 and S2, and S1 and S3 are configured to transmit traffic coming from VLANs 10, 20, 30, and 99 across the network. **This network could not function without VLAN trunks.**

VLANs in a Multi-Switched Environment

VLAN Trunks (cont.)

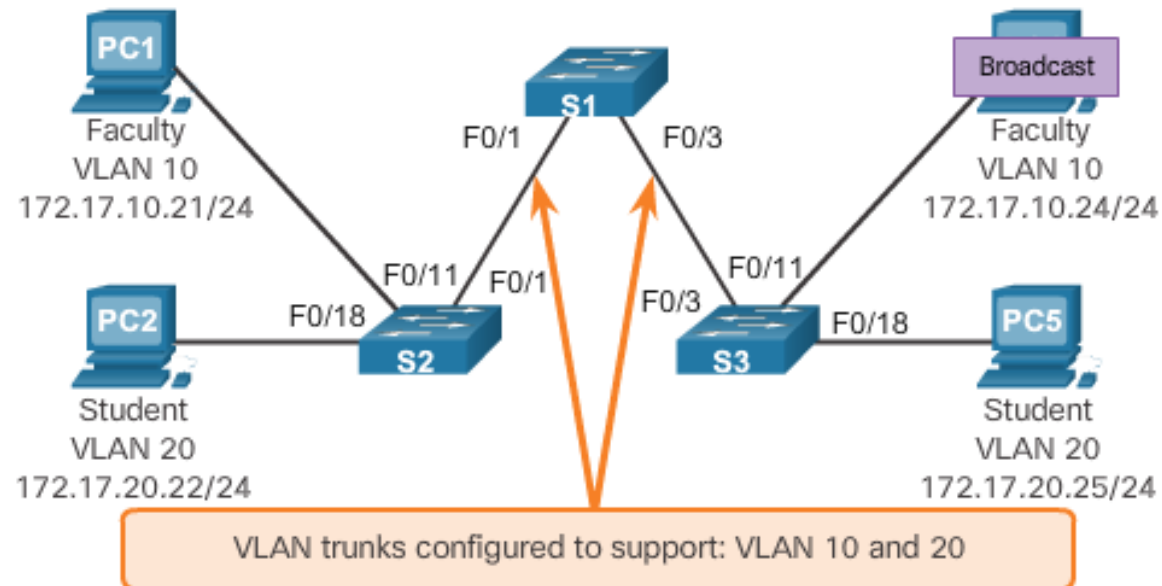
- A VLAN trunk is a point-to-point link that **carries more than one VLAN**.
- A VLAN trunk is usually established between switches so same-VLAN devices can communicate, even if physically connected to different switches.
- **Cisco IOS supports IEEE802.1Q, a popular VLAN trunk protocol.**
- Trunk ports mark frames with unique identifying tags (802.1Q)— as they move between switches.

VLANs in a Multi-Switched Environment

Controlling Broadcast Domains with VLANs

With VLAN Segmentation

PC1 sends out a local Layer 2 broadcast. The switches forward the broadcast frame only out ports configured for VLAN10.



VLANs in a Multi-Switched Environment

Controlling Broadcast Domains with VLANs

- **VLANs can be used to limit the reach of broadcast frames.**
- A VLAN is a broadcast domain of its own.
- A broadcast frame sent by a device in a specific VLAN is forwarded within that VLAN only.
- VLANs help control the reach of broadcast frames and their impact in the network.

VLANs in a Multi-Switched Environment

Native VLANs and 802.1Q Tagging

- Control traffic sent on the native VLAN should not be tagged.
- Frames received untagged, remain untagged and are placed in the native VLAN when forwarded.
- In Cisco switches, the native VLAN is VLAN 1, by default.

VLAN Ranges on Catalyst Switches (cont.)

- Normal Range VLANs

```
Switch# show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gi0/1, Gi0/2
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

VLAN Assignment

Creating a VLAN

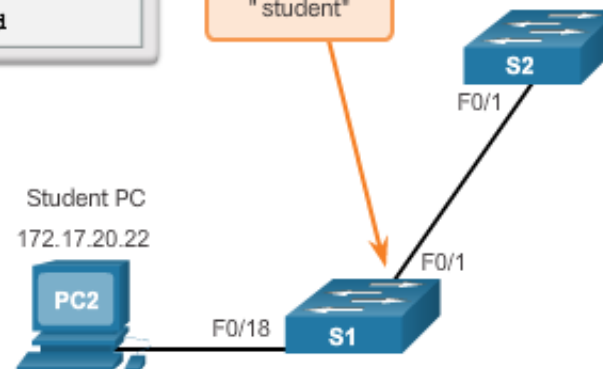
Cisco Switch IOS Commands

Enter global configuration mode.	S1# configure terminal
Create a VLAN with a valid id number.	S1(config)# vlan vlan-id
Specify a unique name to identify the VLAN.	S1(config-vlan)# name vlan-name
Return to the privileged EXEC mode.	S1(config-vlan)# end

Sample Configuration

```
S1# configure terminal
S1(config)# vlan 20
S1(config-vlan)# name student
S1(config-vlan)# end
```

Switch S1:
VLAN 20
"student"



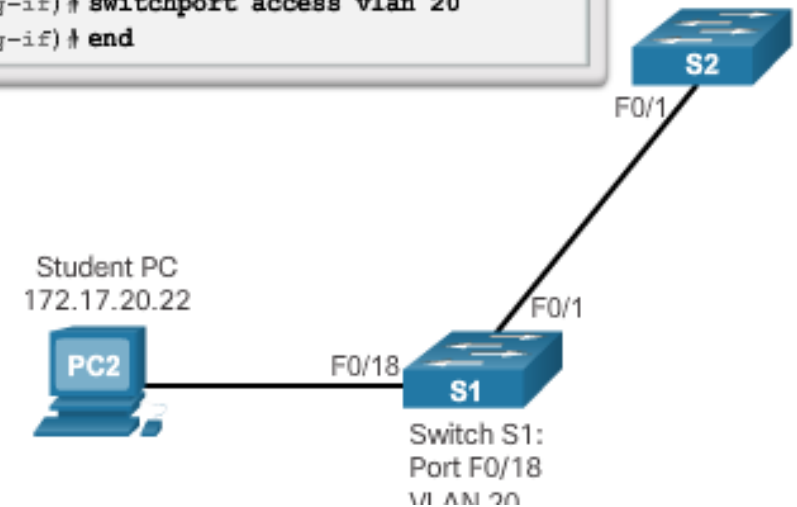
VLAN Assignment

Assigning Ports to VLANs

Cisco Switch IOS Commands

Enter global configuration mode.	<code>S1# configure terminal</code>
Enter interface configuration mode.	<code>S1(config)# interface interface_id</code>
Set the port to access mode.	<code>S1(config-if)# switchport mode access</code>
Assign the port to a VLAN.	<code>S1(config-if)# switchport access vlan vlan_id</code>
Return to the privileged EXEC mode.	<code>S1(config-if)# end</code>

```
s1# configure terminal
s1(config)# interface F0/18
s1(config-if)# switchport mode access
s1(config-if)# switchport access vlan 20
s1(config-if)# end
```



VLAN Assignment

Changing VLAN Port Membership

- Remove VLAN Assignment

Cisco Switch IOS Commands	
Enter global configuration mode.	S1# configure terminal
Remove the VLAN assignment from the port.	S1(config-if)# no switchport access vlan
Return to the privileged EXEC mode.	S1(config-if)# end

- Interface F0/18 was previously assigned to VLAN 20 which is still active, F0/18 reset to VLAN1

```
S1(config)# int F0/18
S1(config-if)# no switchport access vlan
S1(config-if)# end
S1# show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gi0/1, Gi0/2
20	student	active	
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

```
S1#
```

Changing VLAN Port Membership (cont.)

Verification

```
S1# sh interfaces F0/18 switchport
Name: F0/18
Switchport: Enabled
Administrative Mode: static access
Operational Mode: down
Administrative Trunking Encapsulation: dot1q
Negotiation of Trunking: Off
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 1 (default)

<output omitted>
```

VLAN Assignment

Changing VLAN Port Membership (cont.)

Assign Port to VLAN

```
S1# config t
S1(config)# interface F0/11
S1(config-if)# switchport mode access
S1(config-if)# switchport access vlan 20
S1(config-if)# end
S1#
S1# show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24, Gi0/1 Gi0/2
20	student	active	F0/11
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

```
S1#
```

VLAN Assignment

Deleting VLANs

```
S1# conf t
S1(config)# no vlan 20
S1(config)# end
S1#
S1# sh vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24, Gi0/1 Gi0/2
1002 fddi-default	act/unsup	
1003 token-ring-default	act/unsup	
1004 fddinet-default	act/unsup	
1005 trnet-default	act/unsup	

```
S1#
```

- The entire vlan.dat file can be deleted using the **delete flash:vlan.dat** privileged EXEC mode command

Configuring IEEE 802.1q Trunk Links

Trunk Configuration

Cisco Switch IOS Commands

Enter global configuration mode.	<code>S1# configure terminal</code>
Enter interface configuration mode.	<code>S1(config)# interface interface_id</code>
Force the link to be a trunk link.	<code>S1(config-if)# switchport mode trunk</code>
Specify a native VLAN for untagged frames.	<code>S1(config-if)# switchport trunk native vlan vlan_id</code>
Specify the list of VLANs to be allowed on the trunk link.	<code>S1(config-if)# switchport trunk allowed vlan vlan-list</code>
Return to the privileged EXEC mode.	<code>S1(config-if)# end</code>

```
S1(config)# interface FastEthernet0/1
S1(config-if)# switchport mode trunk
S1(config-if)# switchport trunk native vlan 99
S1(config-if)# switchport trunk allowed vlan 10,20,30,99
S1(config-if)# end
```

Resetting the Trunk to Default State

Resetting Configured Values on Trunk Links

Cisco Switch IOS Commands	
Enter global configuration mode.	S1# configure terminal
Enter interface configuration mode.	S1(config)# interface <i>interface_id</i>
Set trunk to allow all VLANs.	S1(config-if)# no switchport trunk allowed vlan
Reset native VLAN to default.	S1(config-if)# no switchport trunk native vlan
Return to the privileged EXEC mode.	S1(config-if)# end

Resetting the Trunk to Default State (cont.)

```
S1(config)# interface f0/1
S1(config-if)# no switchport trunk allowed vlan
S1(config-if)# no switchport trunk native vlan
S1(config-if)# end
S1# show interfaces f0/1 switchport
Name: Fa0/1
Switchport: Enabled
Administrative Mode: trunk
Operational Mode: trunk
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: dot1q
Negotiation of Trunking: On
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 1 (default)
Administrative Native VLAN tagging: enabled
<output omitted>
Administrative private-vlan trunk mappings: none
Operational private-vlan: none
Trunking VLANs Enabled: ALL
Pruning VLANs Enabled: 2-1001
<output omitted>
```

Return Port to Access Mode

```
S1(config)# interface f0/1
S1(config-if)# switchport mode access
S1(config-if)# end
S1# show interfaces f0/1 switchport
Name: Fa0/1
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: native
Negotiation of Trunking: Off
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 1 (default)
Administrative Native VLAN tagging: enabled
<output omitted>
```

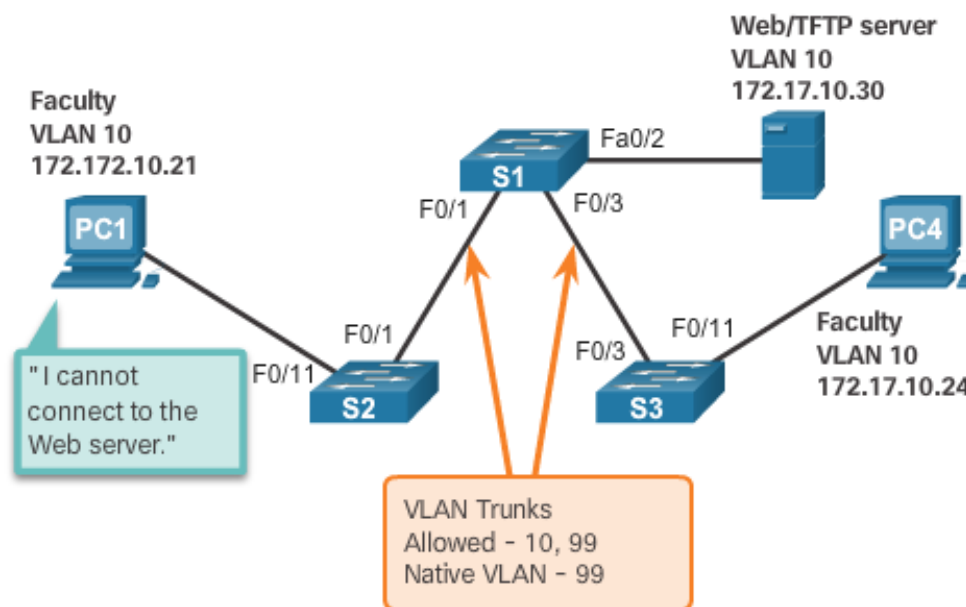
VLAN Trunks

Verifying Trunk Configuration

```
S1(config)# interface f0/1
S1(config-if)# switchport mode trunk
S1(config-if)# switchport trunk native vlan 99
S1(config-if)# end
S1# show interfaces f0/1 switchport
Name: Fa0/1
Switchport: Enabled
Administrative Mode: trunk
Operational Mode: trunk
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: dot1q
Negotiation of Trunking: On
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 99 (VLAN0099)
Administrative Native VLAN tagging: enabled
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk Native VLAN tagging: enabled
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk associations: none
Administrative private-vlan trunk mappings: none
Operational private-vlan: none
Trunking VLANs Enabled: ALL
Pruning VLANs Enabled: 2-1001
<output omitted>
```

IP Addressing Issues with VLANs

- It is a common practice to associate a VLAN with an IP network.
- **Because different IP networks only communicate through a router, all devices within a VLAN must be part of the same IP network to communicate.**
- The figure displays that PC1 cannot communicate to the server because it has a wrong IP address configured.



Common Problems with Trunks

- Trunking issues are usually associated with incorrect configurations.
- The most common type of trunk configuration errors are:
 - Native VLAN mismatches (To solve a native VLAN mismatch, configure the native VLAN to be the same VLAN on both sides of the link.)
 - Trunk mode mismatches
 - (under the interface you wish to use trunking on e.g. interface fa0/1: use the command **Switch(config-if)#switchport mode trunk**
 - This puts the interface into permanent trunking mode and negotiates to convert the link into a trunk link
 - Allowed VLANs on trunks
- If a trunk problem is detected, the best practice guidelines recommend to troubleshoot in the order shown above.

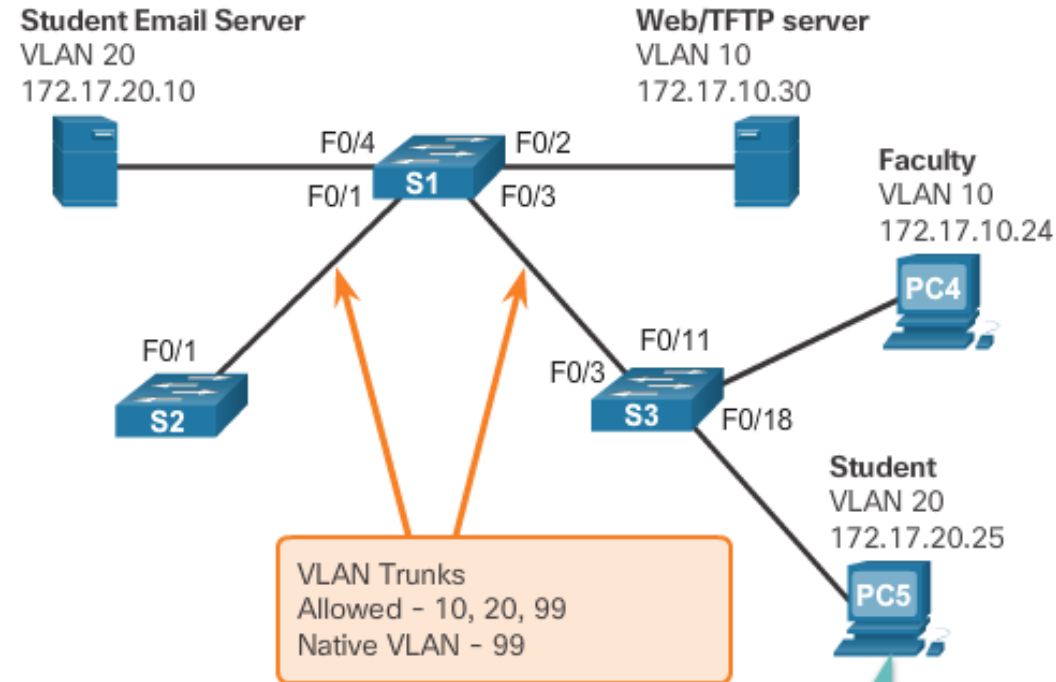
Common Problems with Trunks (cont.)

Problem	Result	Example
Native VLAN Mismatches	Poses a security risk and creates unintended results.	For example, one port is defined as VLAN 99 and the other is defined as VLAN 100.
Trunk Mode Mismatches	Causes loss of network connectivity.	For example, both local and peer switchport modes are configured as dynamic auto.
Allowed VLANs on Trunks	Causes unexpected traffic or no traffic to be sent over the trunk.	The list of allowed VLANs does not support current VLAN trunking requirements.

Troubleshoot VLANs and Trunks

Incorrect VLAN List

Scenario Topology



Output from Switch S1

```
S1# show interfaces trunk
Port      Mode      Encapsulation  Status  Native vlan
Fa0/1     on        802.1q         trunking  99
Fa0/3     on        802.1q         trunking  99
Port      Vlans allowed on trunk
Fa0/1     10,99
Fa0/3     10,99
...
S1#
```

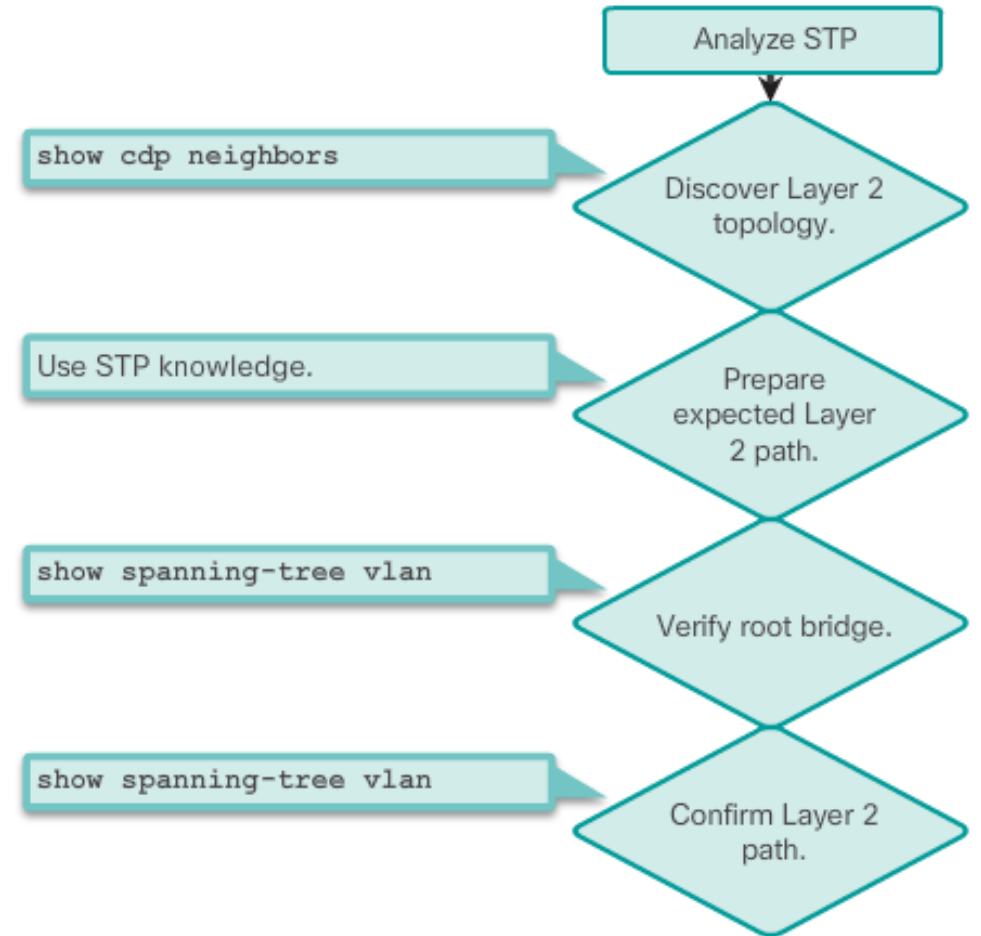
"I cannot connect to the student email server."

Incorrect VLAN List (cont.)

- VLANs must be allowed in the trunk before their frames can be transmitted across the link.
- Use the **switchport trunk allowed vlan** command to specify which VLANs are allowed in a trunk link.
- Use the **show interfaces trunk** command to ensure the correct VLANs are permitted in a trunk.

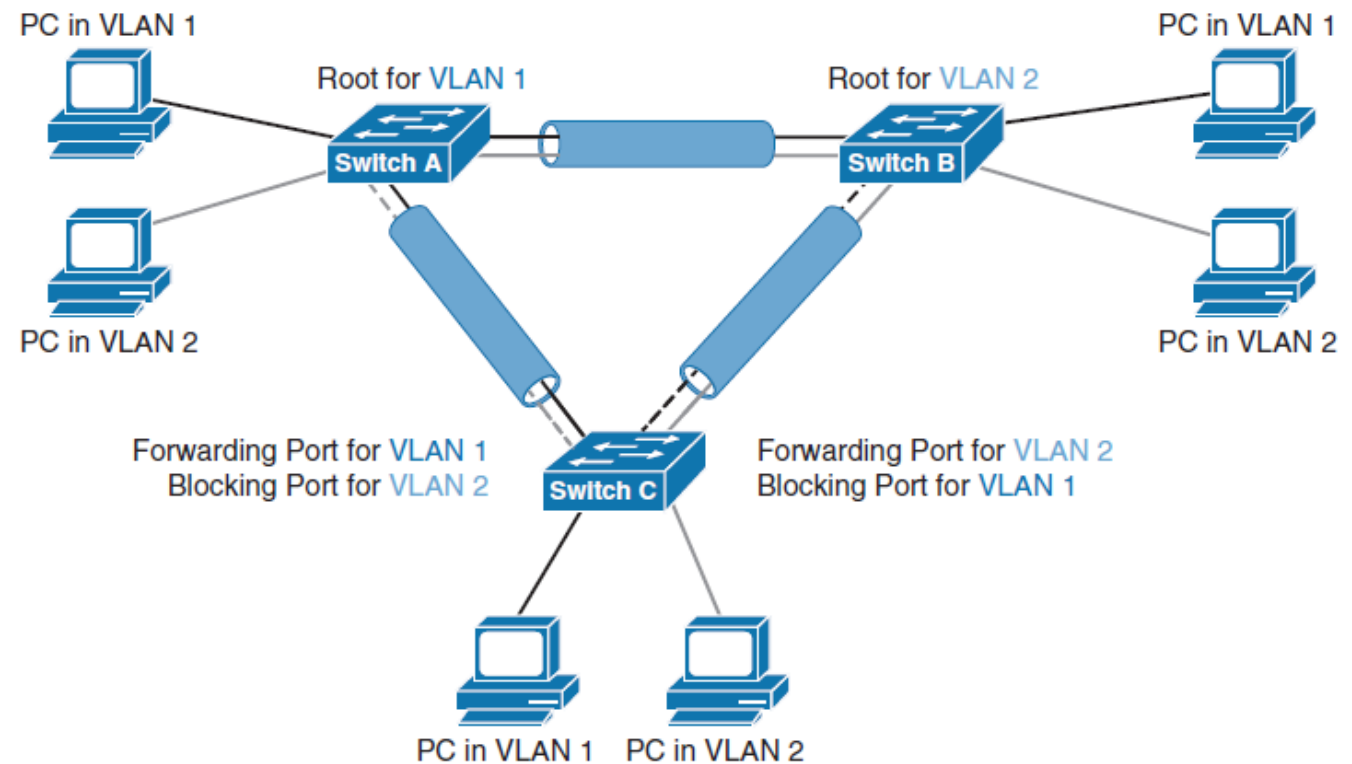
STP Configuration Issues

- Analyzing the STP Topology
- Use **show cdp neighbors** command to discover topology (more info slide 27)
- Use STP knowledge to determine the root switch
- Use **show spanning-tree vlan** to verify which switch is the root and port status (forwarding or blocking).
 - Example-to find out all details for spanning tree for vlan 10 use command:
 - **S2#show spanning-tree vlan 10**



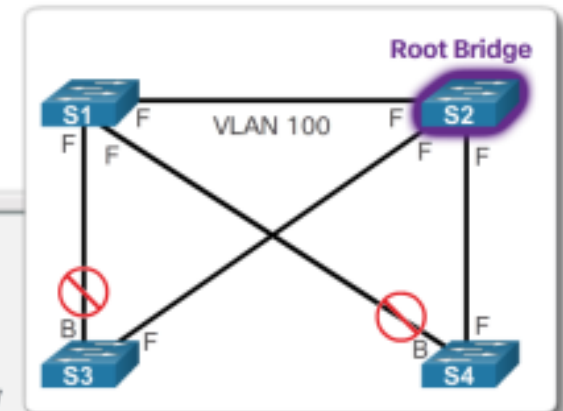
Expected Topology versus Actual Topology

- Troubleshooting consists of comparing the actual state of the network against the expected state of the network and spotting the differences.
- E.g **Per-VLAN STP Plus** (PVST+) is a Cisco implementation of STP that provides a separate spanning-tree instance for each configured VLAN in the network.



Overview of STP Status

- Use the **show spanning-tree** command without specifying any additional options provides a quick overview of the status of STP for all VLANs that are defined on a switch.



- Use the **show spanning-tree vlan** **vlan_id** command to get STP information for a particular VLAN.

```
S1# show spanning-tree vlan 100
```

```
VLAN0100
```

```
Spanning tree enabled protocol rstp
```

```
Root ID    Priority    28772
           Address    0000.0c9f.3127
           Cost      2
           Port      88 (TenGigabit9/1)
           Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Bridge ID   Priority    28772 (priority 28672 sys-id-ext 100)
           Address    0000.0cab.3724
           Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
           Aging Time 300
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Gi3/1	Desg	FWD	4	128.72	P2p
Gi3/2	Desg	FWD	4	128.80	P2p
Te9/1	Root	FWD	2	128.88	P2p

Device Discovery with CDP

- CDP Overview
 - Cisco Discovery Protocol
 - Neighbor discovery of physically connected Cisco devices
- Configure and Verify CDP
 - **show cdp neighbors**
- Discover Devices Using CDP
 - Device identifiers - The host name of the neighbor device
 - Port identifier - The name of the local and remote port
 - Capabilities list - Whether the device is a router or a switch
 - Platform - The hardware platform of the device



Device Discovery with CDP(continued...)

- CDP Overview
 - Cisco Discovery Protocol
 - Neighbor discovery of physically connected Cisco devices

```
Switch0#show cdp neighbors
```

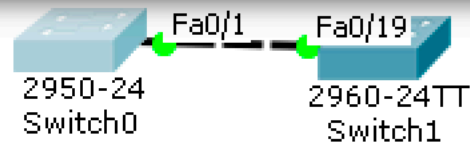
```
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge  
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
```

Device ID	Local Intrfce	Holdtme	Capability	Platform	Port ID
Switch1	Fas 0/1	172	S	2960	Fas 0/19

```
Switch0#
```

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ext 1)



Changing the STP Mode to RSTP

- Rapid PVST+ is the Cisco implementation of per-VLAN RSTP.
- RSTP can achieve **much faster convergence. The convergence time for RSTP is much shorter than for STP. The entire convergence happens at the speed of BPDU transmission.**
- How to enable rapid spanning tree (i.e. it enables 802.1w based Rapid PVST+)

```
SW1(config)# spanning-tree mode rapid-pvst
```

```
SW2(config)# spanning-tree mode rapid-pvst
```

```
SW3(config)# spanning-tree mode rapid-pvst
```

Summary

- STP is a Layer 2 protocol that ensures that there is only one logical path between all destinations on the network by intentionally blocking redundant paths that could cause a loop.
- STP sends BPDU frames for communication between switches. One switch is elected as the root bridge for each instance of spanning tree. An administrator can control this election by changing the bridge priority. Root bridges can be configured to enable spanning tree load balancing by VLAN or by a group of VLANs, depending on the spanning tree protocol used. STP then assigns a port role to each participating port using a path cost. The root path cost is equal to the sum of all the port costs along the path to the root bridge. A port cost is automatically assigned to each port; however, it can also be manually configured. Paths with the lowest cost become preferred, and all other redundant paths are blocked.
- PVST+ is the default configuration of IEEE 802.1D on Cisco switches. It runs one instance of STP for each VLAN. A newer, faster-converging spanning tree protocol, RSTP, can be implemented on Cisco switches on a per-VLAN basis in the form of Rapid PVST+.